

CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · PYQ SOLUTIONS

Chapter 13: Section and Mid-Point Formula

Step-by-step solutions to the previous year questions

The question numbers match the PYQ question sheet.

1 Centroid of a triangle

1. Triangle ABC has vertices $A(0, 6)$, $B(4, -4)$ and $C(-4, -2)$. D is the point where AC crosses the x -axis. (a) Find the coordinates of D . (b) Find the centroid of $\triangle ABC$. (c) D divides AC in the ratio $k : 1$; find k . (d) Find the equation of the line BD . **ICSE 2024, 4 marks**

SOLUTION

(c) Let D divide AC in the ratio $k : 1$. Its y -coordinate is 0: $\frac{-2k + 6}{k + 1} = 0$, so $k = 3$.

(a) Then $x = \frac{3(-4) + 0}{3 + 1} = -3$, so $D = (-3, 0)$.

(b) Centroid = $\left(\frac{0 + 4 - 4}{3}, \frac{6 - 4 - 2}{3}\right) = (0, 0)$

(d) Slope of $BD = \frac{0 - (-4)}{-3 - 4} = -\frac{4}{7}$. Through D : $y = -\frac{4}{7}(x + 3)$, so $7y = -4x - 12$.

Answer: (a) $D(-3, 0)$ (b) $(0, 0)$ (c) $k = 3$ (d) $4x + 7y + 12 = 0$

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